

慢性压迫损伤后大鼠坐骨神经 单细胞转录组图谱与细胞间通讯分析

scRNA-seq generates a molecular map of emerging cell subtypes after sciatic nerve injury in rats

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关键词：单细胞RNA测序 · 慢性压迫损伤（CCI） · 施万细胞 · 配体-受体相互作用 · 神经病理性疼痛

1. 研究背景

外周神经损伤与神经病理性疼痛

- 慢性神经病理性疼痛影响全球 **7-10%** 的人口，现有药物治疗效果有限
- 外周神经损伤后，神经微环境发生剧烈变化——细胞浸润、增殖、通讯网络重构
- 传统观点聚焦于神经元和施万细胞，但 **非神经细胞**（成纤维细胞、免疫细胞、血管细胞）在疼痛发生和维持中的作用被严重低估

研究空白

- 缺乏对外周神经不同解剖分区（神经外膜/束膜/内膜）细胞异质性的系统理解
- 细胞间配体-受体通讯网络在损伤后的动态变化尚未明确
- 谁在“发信号”（配体来源细胞）？谁在“接收信号”（受体表达细胞）？

核心问题：坐骨神经在正常和慢性压迫损伤 (Chronic Constriction Injury, CCI) 后，其细胞组成如何变化？不同细胞类型之间通过哪些配体-受体相互作用进行通讯，从而驱动神经病理性疼痛？

研究策略

- 建立大鼠CCI模型（慢性压迫损伤），采集未受损、3天、12天、60天四个时间点的坐骨神经
- 利用 **10X Genomics** 单细胞转录组测序，鉴定所有细胞类型及其亚群
- 通过 **SingleCellSignalR** 挖掘配体-受体相互作用网络

2. 技术路线



实验样本设计

1. **模型**: Sprague Dawley大鼠, CCI手术 (4根铬肠线结扎坐骨神经, 1mm间距) vs 正常对照
2. **时间点**: 未受损 (Naive) + 损伤后第3天、12天、60天
3. **测序技术**: **10X Genomics Chromium Single Cell 3' Kit v2** (基于微流控油包水液滴技术)
4. **组织准备**: 心脏灌注PBS → 坐骨神经解剖 → 酶消化 → 40μm滤网过滤
5. **空间验证**: RNAscope 原位杂交技术验证关键标记物在组织切片中的真实空间分布

数据分析流程

1. **比对**: CellRanger v3.0.0 → 大鼠基因组 (Rn6)
2. **质控**: 保留200-5500基因且线粒体比例<10%的细胞
3. **标准化**: SCTransform (正则化负二项回归)
4. **批次校正**: Harmony (整合不同时间点和生物学重复)
5. **聚类**: 30个主成分, 分辨率0.8, Leiden聚类
6. **细胞注释**: 基于已知标志基因的差异表达分析

Figure 1: 正常与受损坐骨神经主要细胞类型的分类

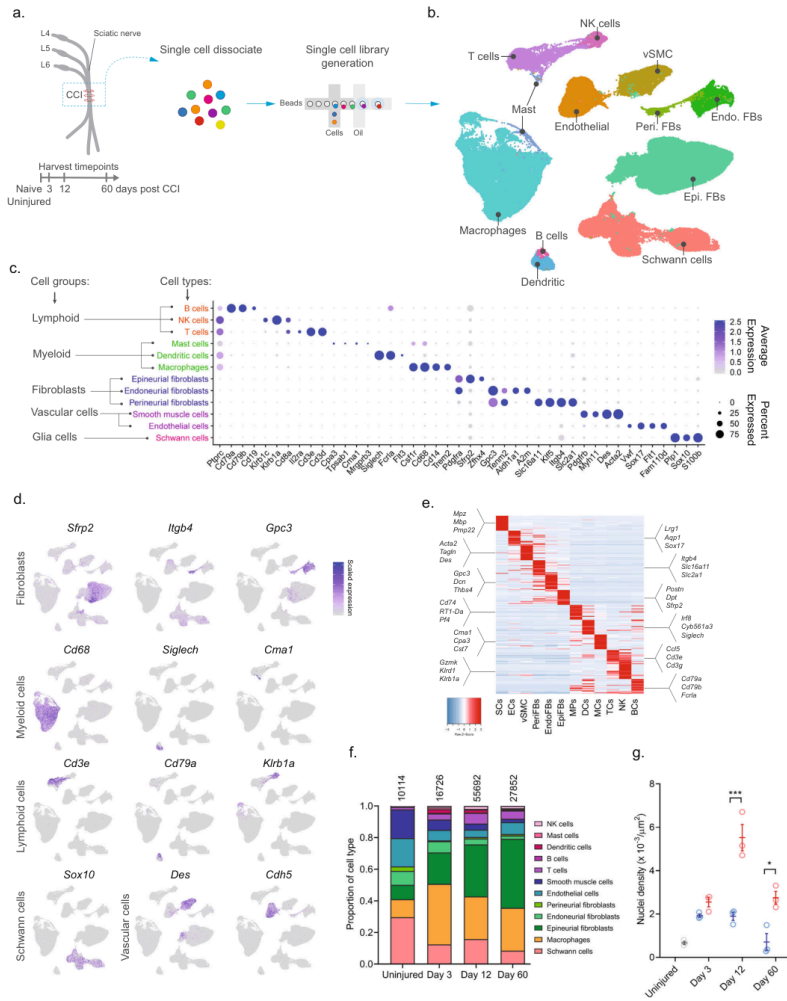


Fig. 1 Classification of major cell types within rat sciatic nerve. **a** Experimental outline of dissociation of sciatic nerve from naive animals and from CCI injured animals at 3, 12, and 60 days post-injury. **b** UMAP plot of clustered cells from all timepoints ($n = 112,521$ cells, $n = 33$ 10X libraries, $n = 18$ animals, $n = 4$ timepoints). **c** Dot plot of scaled expression of genes differentiating major cell types and classification of their group. **d** Feature plot of cell type markers. Cells plotted in random order. **e** Heat map displaying scaled expression of top 20 differentially expressed genes among major cell types using cluster average expression values. **f** Proportion of cell types in scOmics data at each timepoint. **g** Nuclei density in nerve tissue at each timepoint quantified by histology (blue = contralateral, red = ipsilateral, gray = uninjured, individual values and mean $\pm$ SD, $n = 5$ technical replicates per animal, $n = 3$ animals, two-way ANOVA with Sidak's multiple comparisons test, *** $p < 0.0001$, * $p < 0.0035$).

大鼠坐骨神经在正常和CCI损伤后的单细胞图谱 (论文原图, Figure 1, PDF第3页)

小图说明:

- a-c:** 统一流形近似与投影 (Uniform Manifold Approximation and Projection, UMAP) 聚类图展示未受损、3天、12天、60天四个时间点的单细胞图谱及合并数据的全局聚类。
- d-e:** 气泡图/热图 (Heat map) 展示各细胞类型及其亚型的标志基因表达情况。
- f-g:** 细胞比例条形图, 展示不同时间点各类细胞的比例变化。

→ **关键结论:** 无监督聚类识别出32个细胞亚群, 归纳为5大细胞群。

- 实验模型:** 采集未受损 (Naive) 以及CCI手术后第3天、12天和60天的大鼠坐骨神经组织, 利用**10X Genomics**技术进行单细胞解离与测序
- 全局细胞图谱:** 无监督聚类共识别出**32个细胞亚群**, 通过经典标记基因归纳为五大主要细胞群: **胶质细胞 (Glia)**、**成纤维细胞 (Fibroblasts)**、**骨髓免疫细胞 (Myeloid)**、**血管细胞 (Vascular)** 和**淋巴免疫细胞 (Lymphoid)**
- 关键技术澄清:** 该外周组织测序中**未捕获到神经元本体**, 因为外周感觉神经元的细胞体位于背根神经节 (Dorsal Root Ganglion, DRG) 中; 本研究聚焦于复杂的神经微环境及非神经细胞
- 损伤动态响应:** 细胞比例条形图与组织学定量证实, 损伤触发了剧烈的细胞群比例重构, 并在第12天和60天引发显著的细胞浸润与增殖, 导致组织内细胞密度大幅跃升

五大细胞群: 胶质细胞 (Glia)、成纤维细胞 (Fibroblasts)、骨髓免疫细胞 (Myeloid)、血管细胞 (Vascular)、淋巴免疫细胞 (Lymphoid)

3. 测出了哪些细胞类型？—— 五大细胞群的32个亚群

细胞大类	具体亚型举例	关键标志基因	亚群数
胶质细胞 (Glia)	有髓施万细胞、Remak施万细胞、修复型施万细胞、分裂期施万细胞	Sox10, Ngfr, Mbp, Plp1	4种亚型
成纤维细胞 (Fibroblasts)	神经外膜FB、束膜FB、内膜FB、炎症型FB、分裂型FB等	Pdgfra, Col1a, Dpp4, Sfrp2, Gpc3	≥9种亚型 (异质性最高)
骨髓免疫细胞 (Myeloid)	常驻巨噬细胞 (5种亚型)、单核细胞、树突状细胞、中性粒细胞	Cd68, Aif1, C1q, Xcr1	8种亚型 (正常) + 9种 (损伤)
血管细胞 (Vascular)	血管内皮细胞、淋巴管内皮细胞、周细胞、血管平滑肌细胞	Pecam1, Cldn5, Lyve1, Rgs5, Acta2	异质性最低, 损伤后亚型变化少
淋巴免疫细胞 (Lymphoid)	T细胞、NK细胞、NKT细胞	Cd3e, Ncr1, Trbc2	正常神经几乎无, 损伤后大量浸润

关键发现：成纤维细胞 (Fibroblast, FB) 是异质性最高的细胞群体，拥有超过9种以上的亚型，包括三个解剖位置特异性的亚群（外膜/束膜/内膜），以及损伤后大量涌现的炎症型成纤维细胞。

4. 施万细胞亚型——外周神经的驻留神经胶质细胞

施万细胞 (Schwann cell, SC) 是坐骨神经微环境中的**驻留神经胶质细胞**，在正常稳态和损伤响应中发挥核心作用。本研究鉴定出4种功能不同的施万细胞亚型：

施万细胞亚型	特征	标志基因	正常占比	损伤后变化
有髓施万细胞 (Myelinating)	包裹大直径轴突，形成髓磷脂鞘	Mbp, Plp1, Mpz, Prx	58.4%	损伤后去分化，转变为修复表型
Remak施万细胞 (Remak)	包裹小直径轴突（Remak束），无髓	Cdh2, Ngfr	36.1%	同样参与修复响应
修复型施万细胞 (Repair)	损伤后特异性涌现，高表达Ngfr，分泌神经营养因子，清理碎片	Ngfr (p75NTR)高表达	几乎无（仅1个细胞）	3天18% → 12天26% → 60天回归
分裂期施万细胞 (Dividing)	损伤高峰期进入细胞周期进行增殖	Mki67, Ccnb1	极少	损伤高峰期大量出现

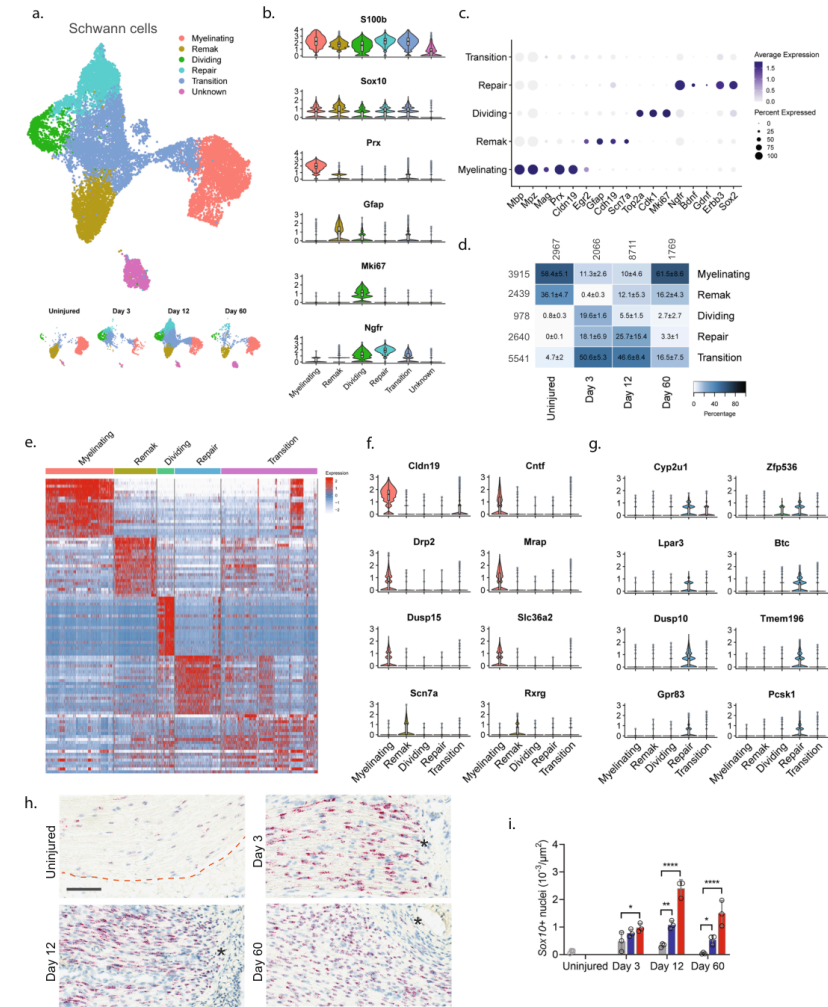
核心发现：正常神经中~94%的施万细胞处于高度特化的有髓/Remak状态。损伤后诱导产生修复型施万细胞（标志物Ngfr/p75NTR），其在3-12天爆发式增长后于60天基本回归基线，提示其为损伤特异性的一过性表型转化。

Figure 2: 正常与受损神经中施万细胞分子亚型的鉴定

施万细胞各亚型的精细分群与标志基因验证 (论文原图, Figure 2, PDF第5页)

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cells (*Gfap*¹⁵), dividing Schwann cells (*Mki67*¹⁶), repair cells (*Ngfr*, also known as *p75NTR*¹⁷), as well as a cluster of cells we termed transition cells (Fig. 2b, c and Supplementary Fig. 5). All the Schwann cell subtypes expressed specific gene signatures further supporting they are subtypes, except for the transition

type, which likely consisted of multiple transitional phenotypes with a temporal signature (Fig. 2d, e).

In naive nerve, 94% of Schwann cells classified as either myelinating (58.4%) or non-myelinating Remak Schwann cells (36.1%) (Fig. 2d). We further identified 0.8% dividing Schwann

小图说明:

1. **a:** UMAP (Uniform Manifold Approximation and Projection) 图展示16,259个施万细胞 (Schwann cell, SC) 在不同时间点的分布, 显示损伤诱导的强烈时间模式变化。
2. **b-c:** 施万细胞亚群精细分群——有髓SC、Remak SC、分裂期SC、修复型SC及过渡态SC (transition cells) 。
3. **d:** 堆叠条形图展示各时间点施万细胞亚型的比例变化——正常神经中94%为有髓/Remak SC, 损伤后修复型SC在3-12天爆发式增长。
4. **e:** 热图展示各施万细胞亚型之间的差异表达基因。
5. **f:** 有髓SC (*Prx*, *Cldn19*, *Cntf*等) 和Remak SC (*Gfap*, *Rxrg*, *Grin2b*等) 的特异性标志基因。
6. **g:** 小提琴图 (Violin plot) 展示修复型SC特异性高表达 *Ngfr*, *Cyp2u1*, *Lpar3*等基因。
7. **h-i:** RNAscope原位杂交 (In Situ Hybridization, ISH) 验证Sox10+施万细胞 (Schwann cell, SC) 密度在不同时间点和解剖分区 (外膜/内膜) 的变化。

1. **亚群精细分群:** 对**16,259**个外周神经胶质细胞——施万细胞进行精细分群, 成功鉴定出有髓施万细胞 (Myelinating)、无髓Remak施万细胞 (Remak)、分裂期施万细胞 (Dividing) 以及修复型施万细胞 (Repair) 等核心亚型
2. **正常组织状态:** 在未受损的正常神经中, 施万细胞呈现高度特化, 约94%的细胞由有髓亚型 (58.4%) 和无髓Remak亚型 (36.1%) 组成
3. **损伤特异性涌现:** 损伤打破了稳态, 诱导产生了一群高表达 *Ngfr* (*p75NTR*) 的修复型施万细胞, 该亚群在正常神经中几乎不存在 (仅1个细胞), 但在损伤后第3天 (18%) 和第12天

5. 关键技术原理与单细胞测序方法

5.1 10X Genomics Chromium 单细胞测序原理

1. **核心原理**：利用微流控芯片将单个细胞与携带独特条形码的凝胶微珠 (Gel Bead in Emulsion, GEM) 包裹在油包水微滴中。每个液滴作为一个独立的"微型反应器"，在液滴内完成细胞裂解、信使核糖核酸 (messenger RNA, mRNA) 捕获和逆转录。
2. **分子标签策略**：每个微珠携带大量相同的细胞条形码和唯一分子标识符 (Unique Molecular Identifier, UMI)，使得测序后能追溯每个转录本来自哪个细胞、哪个原始mRNA分子。
3. **为什么没测出神经元**？坐骨神经中只有轴突缺乏细胞核——轴突mRNA总量极低，标准10X单细胞测序只能捕获有核细胞。神经元胞体位于背根神经节 (Dorsal Root Ganglion, DRG) 中。

5.2 scTransform 标准化原理

1. 使用正则化负二项回归模型对UMI计数进行建模，直接校正测序深度和基因长度的系统性偏差，保留生物学变异。
2. 比传统LogNormalize方法更有效地消除测序深度带来的影响。

5.3 Harmony 批次校正原理

1. 通过"软聚类 (Soft k-means) → 计算数据集特异性偏移 → 线性校正 → 迭代收敛"的流程，消除不同时间点之间的技术偏差。
2. 校正后细胞按其生物学本质（基因表达谱）而非实验来源聚类。

整体分析框架： "10X Genomics单细胞测序" (捕获有核细胞转录组) → "Seurat v3 + Harmony" (质控、标准化、批次校正、聚类) → "SingleCellSignalR" (配体-受体通讯预测) → "RNAscope" (空间验证)。

6. 细胞间通讯分析工具：SingleCellSignalR

6.1 工具概述

1. **软件包**: SingleCellSignalR (R语言, v1.0)
2. **开发者**: Cabello-Aguilar et al., 2020, *Nucleic Acids Research* (PMID: 32246835)
3. **GitHub**: <https://github.com/SCA-IRCM/SingleCellSignalR>

6.2 基本原理——配体-受体 (LR) 共表达分析

1. **核心逻辑**: 基于"配体-受体共表达"原理——如果细胞A表达配体L, 细胞B表达受体R, 则推测A和B之间可能存在通过该配体-受体 (Ligand-Receptor, LR) 对的旁分泌通讯。
2. **配体-受体数据库来源**: 利用SingleCellSignalR框架内设定的一系列**经过人工筛选与整理的公共数据库集 (a set of curated databases)**, 共包含**3251对**已知配体-受体对, 涵盖生长因子、细胞因子、趋化因子、细胞外基质成分等多种信号分子。
3. **LR Score计算**: 对于每对预测的LR相互作用, 计算一个相互作用评分, 评分越高表示该通讯通路的可靠性越高。本研究使用的阈值: 正常神经LR > 0.05, 损伤后LR > 0.5。
4. **输入**: 已经注释好的Seurat对象 (包含细胞类型标签的标准化表达矩阵)。
5. **输出**: 细胞类型之间的配体-受体相互作用矩阵、弦图 (Circos plot)、热图 (Heat map) 等可视化结果。

SingleCellSignalR的核心特点: 提供了从单细胞数据直接推断细胞间通讯的一站式解决方案, 内置LR数据库和评分系统。其基本原理是"配体-受体共表达", 即通过配体细胞和受体细胞之间的表达匹配来推断可能的细胞间通讯。

Figure 8: 正常未受损外周神经内的配体-受体通讯网络

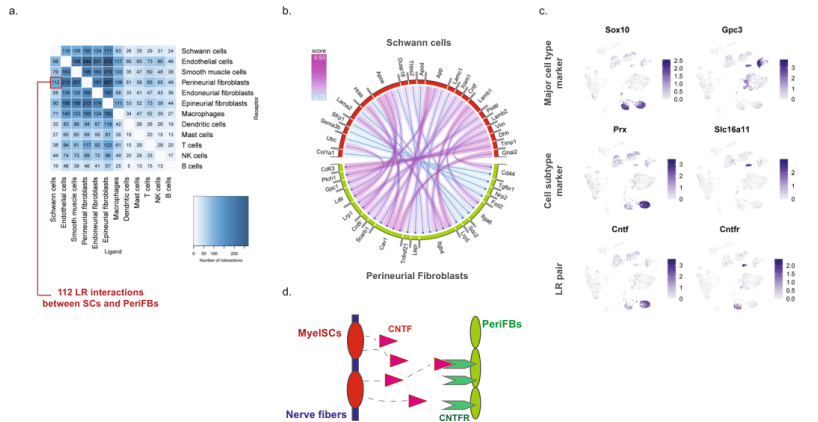


Fig. 8 Ligand-receptor interactions in naive nerve. **a** Number of LR pairs between indicated cell types identified using SingleCellSignalR in naive nerve (LR > 0.05). **b** Circosplot of top interactions signaling from naive Schwann cells (ligand) to naive perineurial fibroblasts (receptor). **c** Feature plots validating cell type markers and identified LR pair in all naive cells. **d** Illustration of signaling between the LR pair, *Cntf*-*Cntfr*, between myelinating Schwann cells and perineurial fibroblasts. The LR score between myelinating Schwann cells and perineurial fibroblasts for *Cntf* and *Cntfr* is 0.84.

Peripheral nerve responds to injury with dramatic swelling and edema, in part due to increased vascularization, cell infiltration and proliferation. In this study, using histology we demonstrate that injury triggers a significant 3–6 fold increase in cell density compared to the contralateral side at 12 and 60 days. Investigation of proliferative markers in the scOmics data, including *Mki67* (Ki67), *Top2a*, and *Cdk11b*, indicate division among all major cell groups and reveal that Schwann cells and epineurial fibroblasts were the most proliferative types with increased densities ranging from 10 to 24 fold. We also identified a significant increase in the density of cell subtypes such as dendritic cells, T cells, and neutrophils which are commonly known to infiltrate tissues, further supporting that some of the increased density is due to infiltration. And last, histological analysis support increased vascularization in both the epi- and endoneurial compartment. Unexpectedly, we also identified a trend, although not significant, of increased cell density on the contralateral side after injury. This was consistent for the total nuclei density across all animals tested and peaked at day 3 or 12 before returning to uninjured levels at day 60.

Analysis of Schwann cells revealed that two subtypes, myelinated and unmyelinated Remak cells, are the prevailing types in naive nerve in agreement with the common consensus on glia in the periphery. In addition, we also identified that about 1% of naive Schwann cells are undergoing cell division suggesting naive nerve continue to replenish and repair even during homeostasis. After injury, however, cell division increased dramatically to 20%, and an injury-specific Schwann cell type emerged transiently and expressed markers of previously characterized Repair cells¹⁹. In fact, during the peak of inflammation, the majority of *Sox10*+ Schwann were either dividing, transitional, or Repair cells, underscoring that the Schwann cell phenotype is highly plastic. Previous studies of Repair cells have suggested the potential of these cells to transition between myelinated and unmyelinated phenotypes via a Repair cell transition, although this has not been

conclusively demonstrated²⁰. In our study, we identified unique and selective markers of the Repair phenotype, which may aid future studies in interrogating the potential for targeting this injury-specific cell type for therapeutic purposes.

Here, we also identified the distinguishing molecular features of multiple subpopulations of myeloid cells in uninjured naive and injured sciatic nerve. We found eight distinct subtypes in naive nerve: *Cma1*+ mast cells, *Xcr1*+/*Clec9a*+ conventional self-renewing tissue-resident dendritic cells, five types of MHC II+ or *C1q*+ macrophages including one dividing and patrolling non-classical *N4a1r*+ monocyte population. Injury was marked by a dramatic increase in the density of myeloid cells, both in the epineurial and endoneurial compartments. Our data further identified the emergence of injury-specific cell subtypes distinguished by molecular features and temporal patterning, and identified recruited neutrophils, *Siglech*+ plasmacytoid dendritic cells, and nine injury subtypes of macrophages. A recent study by Ydens et al. report that the majority of endoneurial and epineurial macrophages are populated by late embryonic precursors that slowly replenish after birth by hematopoietic stem cells derived precursors suggesting this origin could remain in adulthood. This is supported by another study by Wang et al., who show by fate mapping, that the PNS is fully populated with macrophages at birth and only 8% of these derive from the yolk sac. This is in stark contrast to CNS microglia which derive almost exclusively from yolk sac progenitors. Using a *Fli3* transgenic reporter line, Wang et al. further identify that 74% of nerve macrophages were derived from hematopoietic stem cell-derived progenitors. This is significantly more than what we find (~30% of nMP4 in naive nerve), but can be explained by differences in nerve harvest technique and how much epineurial tissue is included in the dissociate. While both reports identify two populations of macrophages distinguished by distinct markers (e.g., *Cd74*, MHC II genes) or (*Clec10a*, *Lyve1^{H1}*), Ydens et al. further show *Clec10a* is restricted to the epineurial compartment. This finding is

正常未受损神经中配体-受体信号通讯机制分析 (论文原图, Figure 8, PDF 第13页)

小图说明:

1. **a:** 热图 (Heat map) 展示正常神经中不同细胞类型间的配体-受体 (Ligand-Receptor, LR) 互作对数 (LR > 0.05)。
2. **b:** 弦图 (Circos plot) 直观展示正常施万细胞 (Schwann cell, SC) (配体) 向束膜成纤维细胞 (Fibroblast, FB) (受体) 的通讯。
3. **c:** 特征图验证细胞类型标记和LR对的表达位置。
4. **d:** 机制图 (Mechanistic diagram)——有髓SC分泌Cntrf通过与束膜FB表面Cntrf结合的信号通路 (Signaling pathway)。

→ 正常稳态中的关键通路: 有髓施万细胞 (Schwann cell, SC)→束膜成纤维细胞 (Fibroblast, FB) (Cntrf→Cntrf, LR score=0.84)。

1. **分析工具与原理:** 采用SingleCellSignalR (基于R语言) 生物信息学框架, 基于"配体-受体共表达"的基本原理, 挖掘含有3251对已知配体-受体对的公共注释数据集, 计算相互作用评分 (LR score)
2. **基线组织稳态:** 在正常未受损神经中, 数千对潜在的旁分泌信号密集分布于胶质、成纤维、血管和骨髓细胞间
3. **重磅常驻通路发现:** 弦图与机制图展示了有髓施万细胞 (Myelinating SCs) 通过释放配体Cntrf, 与解剖学相邻的束膜成纤维细胞 (Perineurial FBs) 表面的受体Cntrf进行直接分子对话 (LR score = 0.84), 该通路在神经发育及炎性痛觉中具有关键功能

→ Cntrf-Cntrf通路: 这是一个在正常稳态中即存在的通讯通路, 揭示施万细胞与成纤维细胞之间的"分子对话"维持着神经束膜的完整性。

Figure 9: 损伤后外周神经内配体-受体通讯网络重构

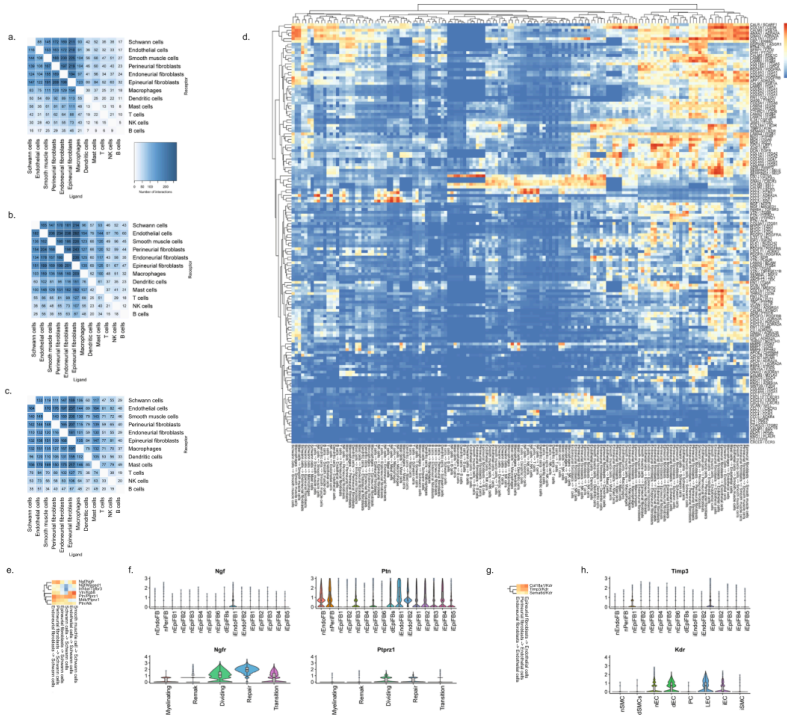


Fig. 9 Ligand-receptor interactions after injury. Number of LR interactions larger than 0.5 between the indicated two cell types after **a** 3 days, **b** 12 days, and **c** 60 days of CCI injury, respectively. **d** Heat map of scaled expression of top 100 most-variable interactions at day 12. **e** Subset of interactions from (**d**) between types of fibroblasts and Schwann cells. **f** Violin plot with boxplot overlay and outliers (dots) of highlighted pairs, *Ngf* - *Ngfr* and *Ptn* - *Ptptr1*, from (**e**). **g** Subset of interactions from (**d**) between types of fibroblasts and vascular cells. **h** Violin plot with boxplot overlay and outliers (dots) of highlighted pair, *Timp3* - *Kdr*, from (**g**).

supported by a recent study in lung tissues where MHC II genes and *Lyve1* distinguished macrophages associated with nerve bundles and vessels, respectively, adding further support to anatomical functional specialization in the peripheral nerve compartment⁸¹. Based on these previously validated markers, our study suggests that the epineurial compartment is populated by three subtypes of self-maintaining macrophages involved in tissue wound repair, while the endoneurial compartment is populated by one subtype of macrophage derived from both embryonic and hematopoietic stem cells lineages, but with a similar phenotype involved in antigen presentation. Further studies are needed to fully delineate the molecular types and their respective functional roles in homeostasis and repair.

In our study, fibroblasts had the most diverse phenotypical heterogeneity. We identified three different anatomical types comprising more than nine types in naive nerve and an additional seven types in injured nerve. Histological analysis demonstrated

robust proliferation and the development of an inflammatory profile among fibroblasts as well as the emergence of many injury triggered epineurial types peaking at day 12. The role of fibroblasts in nociception remains not fully understood. A previous study by Carr et al. investigated nerve injury and proposed that endoneurial fibroblasts, also termed mesenchymal cells, possess precursor potential for bone and dermal repair⁶. The major cell type with the least heterogeneity were vascular cells. This cell group also displayed minimal emergence of distinct subtypes after injury although types displayed robust DEGs after, but not before, injury. This is not surprising since the vasculature may be uniquely equipped with mechanisms to quickly deal with neurogenic inflammation.

Lymphoid cells displayed a unique profile. While the uninjured nerve was essentially devoid of lymphoid cells, many different subtypes infiltrated after injury, which was also supported by histological evidence. Given the known challenges in phenotyping

CCI损伤后神经中配体-受体信号通讯机制分析 (论文原图, Figure 9, PDF第14页)

小图说明:

1. **a-c:** 热图 (Heat map) 展示损伤后细胞类型间的配体-受体 (Ligand-Receptor, LR) 互作对数显著增加 (LR > 0.5) 。
2. **d:** 热图 (Heat map) 展示第12天变异性 (Variability) 最大的前100个相互作用的缩放表达。
3. **e:** 热图 (Heat map) 展示成纤维细胞 (Fibroblast, FB) 和施万细胞 (Schwann cell, SC) 之间相互作用的子集 (Subset)。
4. **f:** 小提琴图 (Violin plot) 展示损伤后*Ngf*在成纤维细胞 (Fibroblast, FB) 中的表达显著上调, *Ngfr*在修复型施万细胞 (Schwann cell, SC) 中的表达也上调; 同时展示*Ptn*在成纤维细胞 (Fibroblast, FB) 中的表达上调, *Ptptr1*在修复型施万细胞 (Schwann cell, SC) 中的表达上调。

→ 损伤诱导的痛觉触发通路: 成纤维细胞 (Fibroblast, FB)→修复型施万细胞 (Schwann cell, SC) (*Ngf*→*Ngfr*) , 以及*Ptn*→*Ptptr1*通路。

1. **损伤痛觉触发通路:** 损伤激活后, 细胞间通讯网络彻底重构。分析表明, 修复型施万细胞表面的神经生长因子受体 (*Ngfr/p75NTR*) 被强烈激活, 其配体直接来源于成纤维细胞高分泌的神经生长因子 (*Ngf*) , 这揭示了成纤维细胞可能是触发和维持损伤后外周神经病理性疼痛的药理学核心干预靶点
2. **第二关键通路:** 成纤维细胞分泌的多效生长因子 (*Ptn*) 与修复型施万细胞表面的受体蛋白酪氨酸磷酸酶ζ1 (*Ptptr1*) 之间的相互作用, 参与损伤后施万细胞的增殖与分化调控
3. **网络重构程度:** 损伤诱导的LR互作网络较正常状态显著扩大, 成纤维细胞成为通讯网络的"枢纽", 连接多种细胞类型

→ **核心结论:** 损伤使细胞间通讯网络从稳态维持模式转变为痛觉信号驱动模式, 成纤维细胞 (Fibroblast, FB) 处于该转变的枢纽位置。成纤维细胞→修复型施万细胞的*Ngf*→*Ngfr*通路是关键痛觉触发通路。

7. 有没有神经细胞相关的细胞间通讯？

施万细胞——既是通讯的发送者也是接收者

在本研究中，施万细胞 (Schwann cell, SC) 作为**唯一的驻留神经胶质细胞**，在配体-受体通讯网络中同时扮演发送者和接收者角色：

通讯角色	配体/受体	通讯伙伴	生理意义
发送者（配体输出）	有髓施万细胞 → 分泌Cntf	→ 束膜成纤维细胞（Cntfr受体）	稳态维持，束膜完整性调控（LR score = 0.84）
接收者（受体输入）	修复型施万细胞 ← Ngfr (p75NTR)	← 内膜成纤维细胞分泌Ngf	损伤后疼痛信号放大
接收者（受体输入）	修复型施万细胞 ← Ptprz1	← 成纤维细胞分泌Ptn	损伤后细胞增殖与分化调控
发送者与接收者	Timp3-Kdr通路	成纤维细胞Timp3 → 血管细胞Kdr (VEGFR2)	损伤诱导的血管新生调控

关键发现：施万细胞与成纤维细胞 (Fibroblast, FB) 之间的双向通讯是神经微环境稳态和损伤后疼痛的核心驱动力。特别是成纤维细胞→修复型施万细胞（Ngf→Ngfr）通路，揭示了一个此前未被充分认识的上游调控机制——成纤维细胞很可能是触发和维持损伤后外周神经病理性疼痛的药理学核心干预靶点。

8. 结论与展望

核心发现

- 首次在单细胞分辨率下系统描绘了大鼠坐骨神经在慢性压迫损伤 (Chronic Constriction Injury, CCI) 损伤后4个时间点的细胞图谱, 鉴定出 **32个细胞亚群**
- 施万细胞 (Schwann cell, SC)** 在损伤后表现出高度可塑性: 有髓→修复型去分化、分裂增殖、一过性修复型爆发
- 成纤维细胞 (Fibroblast, FB)** 是异质性最高的细胞群体 (≥ 9 个正常亚型 + 损伤特异性亚型), 在通讯网络中处于枢纽位置
- SingleCellSignalR**分析揭示: 正常中有Cntf→Cntfr (稳态维持), 损伤后有Ngf→Ngfr和Ptn→Ptprz1 (痛觉触发)

临床转化潜力

- 为开发靶向成纤维细胞-施万细胞通讯的镇痛药物提供了分子基础
- 修复型施万细胞的标志物 (Cyp2u1、Lpar3) 可作为神经病理性疼痛的潜在生物标志物

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